ERIC M. TANG, PH.D.

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SUMMARY

Research scientist specializing in image processing, computer vision, and machine learning for medical imaging. Eight years of experience spanning optical hardware, real-time acquisition software, and deep learning — from prototype systems on the optical bench to devices used in the clinic. Comfortable owning the full stack: sensors and control systems, multithreaded C++ acquisition pipelines, and the models that turn raw signal into clinical insight.

TECHNICAL SKILLS

Programming Languages & Tools (8 years): Python, C/C++ (multithreaded, real-time), MATLAB, Java; Visual Studio, VS Code, Qt Creator, LabVIEW

Machine Learning & AI (6 years): Deep learning (PyTorch/LibTorch, TensorFlow), CNNs (U-Net, YOLO), GANs; real-time object detection, classification, segmentation, and tracking; dataset curation and annotation QC; model training, evaluation, and optimization

Model Optimization & Deployment: PyTorch model export via ONNX, TensorRT optimization for latency-constrained real-time inference on embedded hardware (NVIDIA Jetson Orin Nano); integration into low-latency C++ runtimes (LibTorch)

Computer Vision (8 years): Geometric computer vision (camera calibration, stereo vision, 3D reconstruction; OpenCV); signal and image processing, denoising, super-resolution, multimodal image fusion

Camera & Imaging Systems (8 years): Multimodal camera and sensor integration (visible-light, near-infrared OCT, photoacoustic); 3D/4D imaging systems; multi-view acquisition; camera sensor characterization; devices from prototype through validation

Hardware (8 years): Control systems, DAQ/FPGA programming, hardware control APIs (cameras, scanners, lasers, sensors), optical system design and optimization

EXPERIENCE

Topcon Healthcare — San Jose, CA May 2023 – Present
Senior Research Scientist

- Manage and optimize real-time image acquisition software (C++) for device control, high-throughput data capture, and flexible customization of imaging workflows via multithreaded programming
- Designed a physics-informed deep learning model (PyTorch) for image super-resolution and denoising, integrating the optical system's PSF and acquisition parameters; achieved a 2× improvement in resolution and signal quality, presented to executives and stakeholders
- Deployed deep learning imaging models to embedded edge hardware by exporting PyTorch models via ONNX and optimizing with TensorRT for real-time inference on an NVIDIA Jetson Orin Nano
- Lead cross-organizational collaborations between vendors and engineering teams to co-develop a novel compact imaging sensor and scale it from development to production, achieving a 2× reduction in device size; contribute hardware integration, software evaluation, and quantitative benchmarking

Diagnostic Imaging & Image-Guided Interventions Lab — Nashville, TN May 2022 – May 2023
Postdoctoral Research Fellow

- Extended and adapted a deep-learning pipeline for 4D surgical-instrument tracking with multi-channel inputs and data augmentation, improving robustness across imaging conditions to 99% sensitivity at 23 fps
- Curated and managed multimodal image datasets for model training, including labeling, annotation quality control, and preprocessing pipelines
- Supported development, training, and real-time deployment of an image denoising framework using multi-scale U-Net architectures, improving image quality by 90% at video rate (22 fps)

Vanderbilt University — Nashville, TN May 2018 – May 2022
Graduate Research Assistant

- Optimized a high-speed multimodal OCT imaging system and managed its acquisition codebase (C++) supporting real-time processing, live preview of reflectance and 3D images, and GPU-accelerated CNN architectures at 4 gigasamples/second
- Integrated an automated deep-learning framework for surgical-instrument tracking using multi-view inputs and a lightweight model (YOLOv4) for real-time acquisition and detection at 120 fps

EDUCATION

Vanderbilt University — Ph.D., Biomedical Engineering 2022

Duke University — B.S., Biomedical Engineering 2018

Duke University — B.A., Computer Science 2018

SELECTED PUBLICATIONS

Tang, E. M., El-Haddad, M. T., Patel, S. N., Tao, Y. K. "Automated instrument-tracking for 4D video-rate imaging of ophthalmic surgical maneuvers." Biomedical Optics Express 13(3), 1471–1484 (2022).

Rico-Jimenez, J. J., Hu, D., **Tang, E. M.**, Oguz, I., Tao, Y. K. "Real-Time OCT Image Denoising Using Self-Fusion Neural Network." Biomedical Optics Express 13(3), 1398–1409 (2022).

Full publication list: Google Scholar

PROJECTS

ericmtang.com ("TangOS") — a personal website with the look and feel of a desktop OS, built from scratch with Next.js, React, and TypeScript.